

Assignment 3

Note: there can be more than one correct answer.

Q1

1) [0.20 points]

Give regular expressions generating the following languages over the alphabet $\Sigma = \{0,1\}$:

- a. $\{w \mid w \text{ begins with a 1 and ends with a 0}\}$
- b. $\{w \mid w \text{ contains at least three 1s}\}$
- c. $\{w \mid w \text{ contains the substring 0101 (i.e., } w = x0101y \text{ for some } x \text{ and } y)\}$
- d. $\{w \mid w \text{ has length at least 3 and its third symbol is a 0}\}$
- e. $\{w \mid w \text{ starts with 0 and has odd length, or starts with 1 and has even length}\}$
- f. $\{w \mid w \text{ doesn't contain the substring 110}\}$
- g. $\{w \mid \text{the length of } w \text{ is at most 5}\}$
- h. $\{w \mid w \text{ is any string except 11 and 111}\}$
- i. $\{w \mid \text{every odd position of } w \text{ is a 1}\}$
- j. $\{w \mid w \text{ contains at least two 0s and at most one 1}\}$
- k. $\{\epsilon, 0\}$
- l. $\{w \mid w \text{ contains an even number of 0s, or contains exactly two 1s}\}$
- m. The empty set
- n. All strings except the empty string

A1

- a) $\{w \mid w \text{ begins with a 1 and ends with a 0}\} \quad R = 1\Sigma^*0$
- b) $\{w \mid w \text{ contains at least three 1s}\} \quad R = \Sigma^*1\Sigma^*1\Sigma^*1\Sigma^*$
- c) $\{w \mid w \text{ contains the substring 0101 (i.e., } w = x0101y \text{ for some } x \text{ and } y)\}$
 $R = \Sigma^*0101\Sigma^*$
- d) $\{w \mid w \text{ has length at least 3 and its third symbol is a 0}\} \quad R = \Sigma\Sigma 0\Sigma^*$
- e) $\{w \mid w \text{ starts with 0 and has odd length, or starts with 1 and has even length}\}$
 $R = 0(\Sigma\Sigma)^* \cup 1\Sigma(\Sigma\Sigma)^*$
- f) $\{w \mid w \text{ doesn't contain the substring 110}\} \quad R = (0 \cup (10))^*1^*$
- g) $\{w \mid \text{the length of } w \text{ is at most 5}\}$
 $R = \epsilon \cup \Sigma \cup \Sigma\Sigma \cup \Sigma\Sigma\Sigma \cup \Sigma\Sigma\Sigma\Sigma \cup \Sigma\Sigma\Sigma\Sigma\Sigma$

h) $\{w \mid w \text{ is any string except } 11 \text{ and } 111\}$ $R = \epsilon \cup 1 \cup 1^*0\Sigma^* \cup 1111\Sigma^*$

i) $\{w \mid \text{every odd position of } w \text{ is a } 1\}$

The correct solution should be $R = (1\Sigma)^*1 \cup (1\Sigma)^+ \cup \epsilon$ (The empty string is included). Because of the ambiguity of the English word "odd", the following answer is also marked as correct.

$$R = (1\Sigma)^*1 \cup (1\Sigma)^+$$

j) $\{w \mid w \text{ contains at least two 0s and at most one 1}\}$

$$R = (00 \cup 000^*1 \cup 010 \cup 100)0^*$$

k) $\{\epsilon, 0\}$ $R = 0 \cup \epsilon$

l) $\{w \mid w \text{ contains an even number of 0s, or contains exactly two 1s}\}$

The correct solution should be: $R = (1^*01^*01^*)^* \cup 0^*10^*10^*$ (The empty string is included because it contains zero 0s, and zero is even). But because of the ambiguity, the following regular expression (the empty string is excluded) is also marked as correct:

$$R = (1^*01^*01^*)^+ \cup 0^*10^*10^*$$

m) The empty set $R = \emptyset$

n) All strings except the empty string $R = (\Sigma)^+$

Q2

2) [0.30 points]

Use the construction proof to convert the following regular languages to NFAs:

- a. $(0 \cup 1)^*010(0 \cup 1)^*$
- b. $((((11)^*(00)) \cup 01)^*$
- c. $\emptyset^*$

A2

a) $(0 \cup 1)^*010(0 \cup 1)^*$

b) $((11)^*(00) \cup 01)^*$

$0 \rightarrow 0 \xrightarrow{0} \odot$

$11 \rightarrow 0 \xrightarrow{1} 0 \xrightarrow{\varepsilon} 0 \xrightarrow{1} \odot$

$00 \rightarrow 0 \xrightarrow{0} 0 \xrightarrow{\varepsilon} 0 \xrightarrow{0} \odot$

$01 \rightarrow 0 \xrightarrow{0} 0 \xrightarrow{\varepsilon} 0 \xrightarrow{1} \odot$

$(11)^* \rightarrow \odot \xrightarrow{\varepsilon} 0 \xrightarrow{1} 0 \xrightarrow{\varepsilon} 0 \xrightarrow{1} \odot$

$(11)^*(00) \rightarrow \odot \xrightarrow{\varepsilon} 0 \xrightarrow{1} 0 \xrightarrow{\varepsilon} 0 \xrightarrow{1} 0 \xrightarrow{\varepsilon} 0 \xrightarrow{0} 0 \xrightarrow{\varepsilon} 0 \xrightarrow{0} \odot$

$((11)^*(00) \cup 01) \rightarrow \odot \xrightarrow{\varepsilon} 0 \xrightarrow{0} 0 \xrightarrow{\varepsilon} 0 \xrightarrow{1} \odot$
 $\rightarrow 0 \xrightarrow{\varepsilon} 0 \xrightarrow{1} 0 \xrightarrow{\varepsilon} 0 \xrightarrow{1} 0 \xrightarrow{\varepsilon} 0 \xrightarrow{0} 0 \xrightarrow{\varepsilon} 0 \xrightarrow{0} \odot$

$((11)^*(00) \cup 01)^* \rightarrow \odot \xrightarrow{\varepsilon} 0 \xrightarrow{0} 0 \xrightarrow{\varepsilon} 0 \xrightarrow{1} \odot$
 $\rightarrow \odot \xrightarrow{\varepsilon} 0 \xrightarrow{1} 0 \xrightarrow{\varepsilon} 0 \xrightarrow{1} 0 \xrightarrow{\varepsilon} 0 \xrightarrow{0} 0 \xrightarrow{\varepsilon} 0 \xrightarrow{0} \odot$

c) $\emptyset^*$

c. $\emptyset^*$

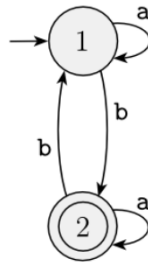
$\emptyset \rightarrow 0$

$\emptyset^* \rightarrow \odot \xrightarrow{\varepsilon} 0$

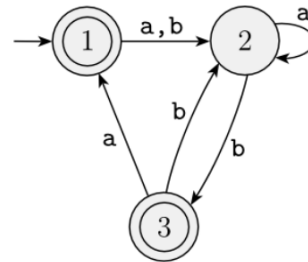
Q3

3) [0.40 points]

Use the construction proof to convert the following finite automata to regular expressions:



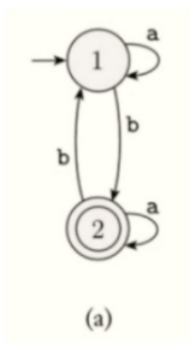
(a)



(b)

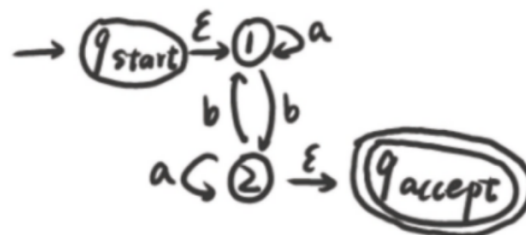
A3

a)

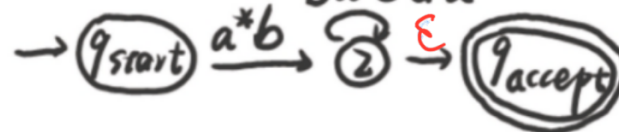


(a)

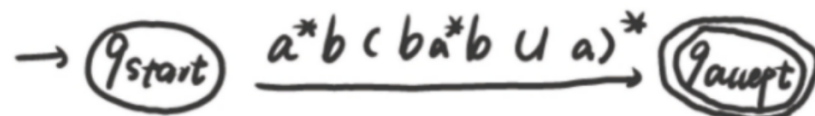
① 2-state DFA $\Rightarrow$ 4-state GNFA



② 4-state GNFA $\Rightarrow$ 3-state GNFA
 $ba^*b \cup a$

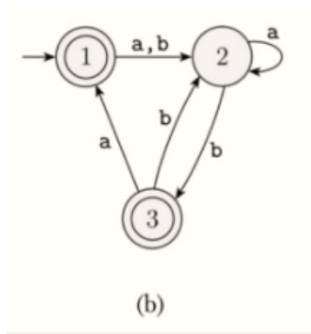


③ 3-state GNFA $\Rightarrow$ 2-state GNFA

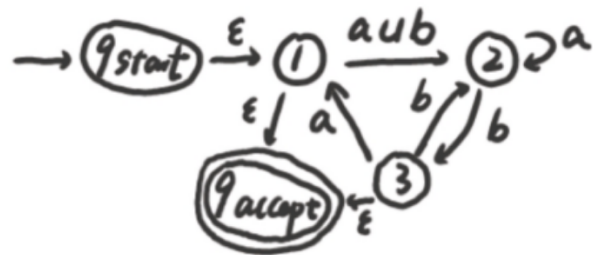


④ regular expression: $a^*b(ba^*b \cup a)^*$

b)



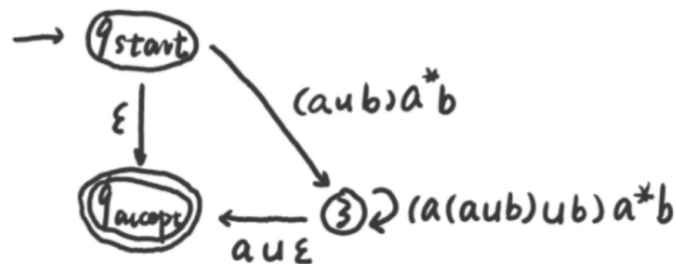
① 3-state DFA $\Rightarrow$ 5-state GNFA



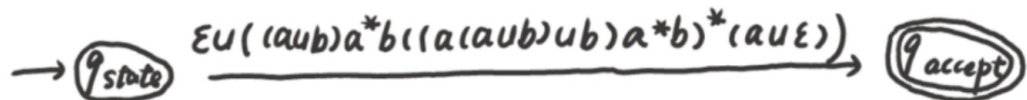
② 5-state GNFA $\Rightarrow$ 4-state GNFA



③ 4-state GNFA $\Rightarrow$ 3-state GNFA



④ 3-state GNFA $\Rightarrow$ 2-state GNFA



⑤ regular expression: $\epsilon|u|(a|b)a^*b((a(a|b)|ub)a^*b)^*(a|u|\epsilon)$

Q4

4) [0.60 points]

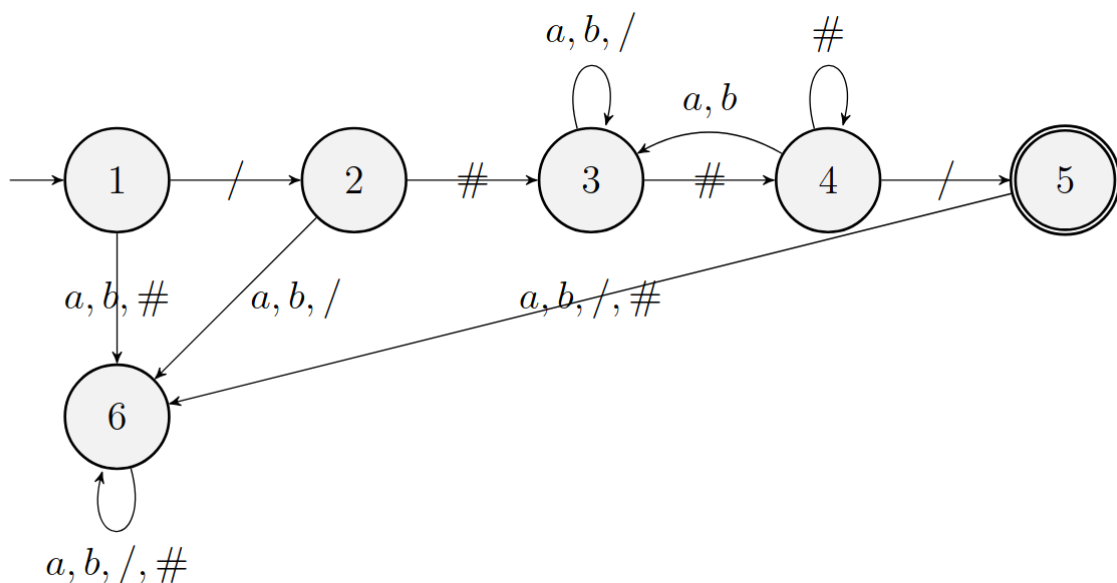
In certain programming languages, comments appear between delimiters such as `/#` and `#/`. Let C be the language of all valid delimited comment strings. A member of C must begin with `/#` and end with `#/` but have no intervening `#/`. For simplicity, assume that the alphabet for C is $\Sigma = \{a, b, /, \#\}$.

- Give a DFA that recognizes C .
- Give a regular expression that generates C .

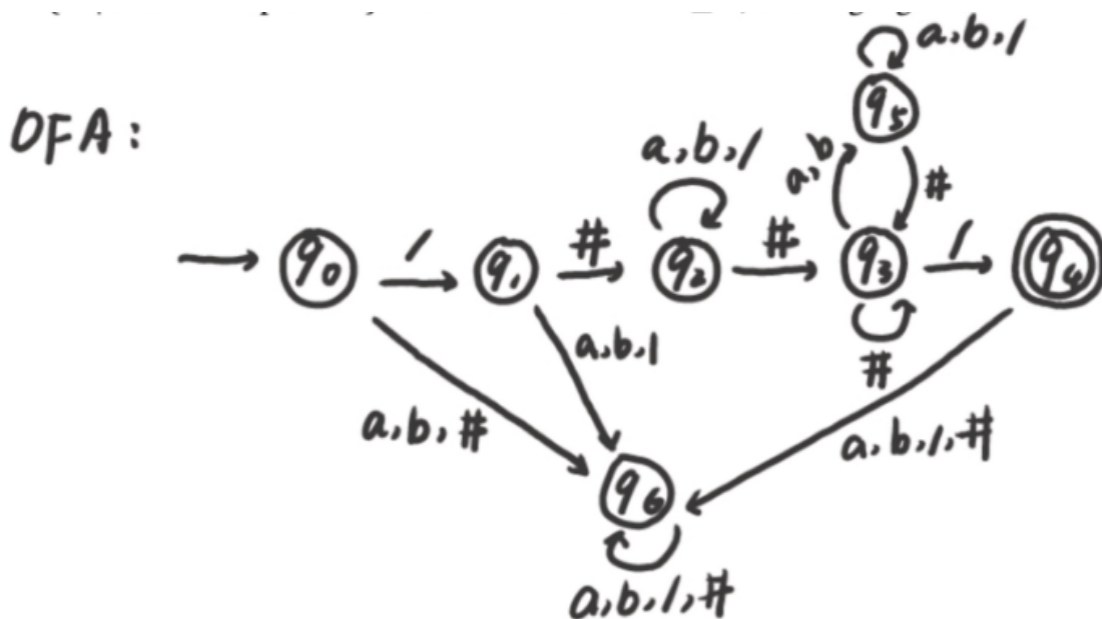
A4

a) Give a DFA that recognizes C .

Version 1,

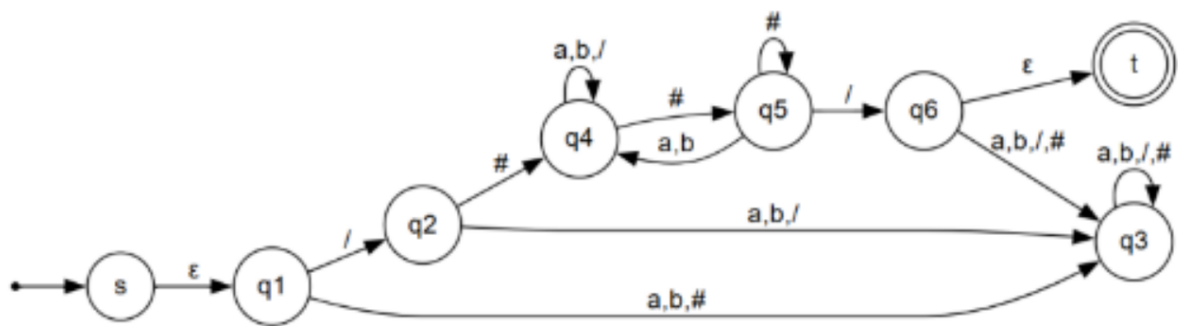


Version 2,

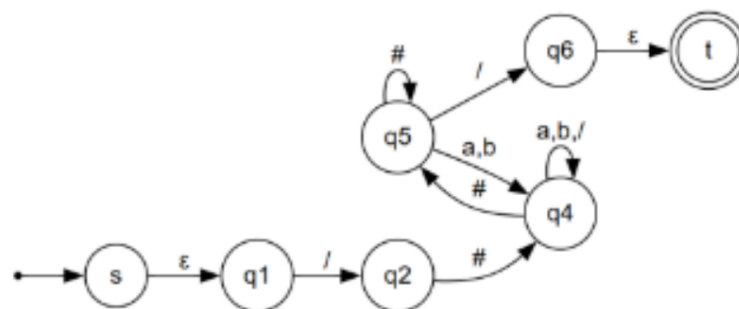


b) Give a regular expression that generates C .

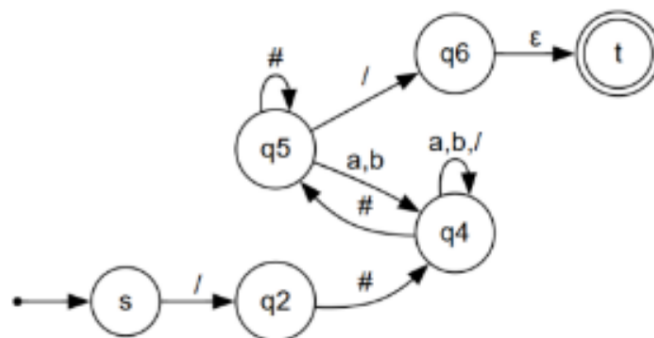
The GNFA that recognizes C is,



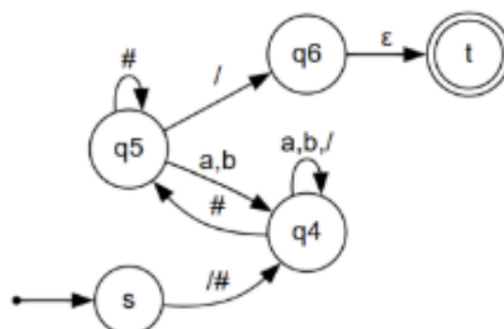
Convert $q3$,



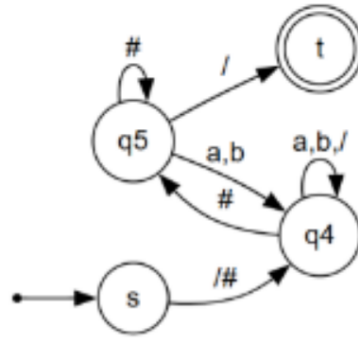
Convert $q1$,



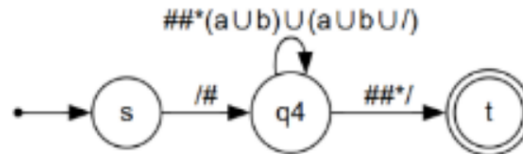
Convert $q2$,



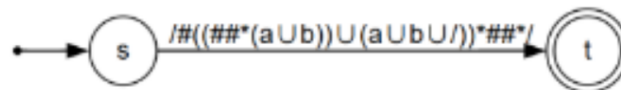
Convert $q6$,



Convert q5,



Convert q4,



Q5

5) [0.5 points]

For any string $w = w_1w_2 \cdots w_n$, the **reverse** of w , written $w^{\mathcal{R}}$, is the string w in reverse order, $w_n \cdots w_2w_1$. For any language A , let $A^{\mathcal{R}} = \{w^{\mathcal{R}} \mid w \in A\}$. Show that if A is regular, so is $A^{\mathcal{R}}$.

A5

if A is regular, then $A = L(R)$ for some regular expression R .

For any regular expression R , there exists a regular expression $R^{\mathcal{R}}$, that $L(R^{\mathcal{R}}) = L(R)^{\mathcal{R}}$

Proof :

- (1): $R = a, a \in \Sigma$ then $R^{\mathcal{R}} = a$
- (2): $R = \varepsilon$ then $R^{\mathcal{R}} = \varepsilon$
- (3): $R = \emptyset$ then $R^{\mathcal{R}} = \emptyset$
- (4): $R = (R_1 \cup R_2)$ then $R^{\mathcal{R}} = (R_2^{\mathcal{R}} \cup R_1^{\mathcal{R}})$
- (5): $R = (R_1 \circ R_2)$ then $R^{\mathcal{R}} = (R_2^{\mathcal{R}} \circ R_1^{\mathcal{R}})$
- (6): $R = (R_1)^*$ then $R^{\mathcal{R}} = (R_1^{\mathcal{R}})^*$

By the definition of regular expression, QED.

Hence, there exists a regular expression $R^{\mathcal{R}}$ that $L(R^{\mathcal{R}}) = A^{\mathcal{R}}$, then $A^{\mathcal{R}}$ is regular.

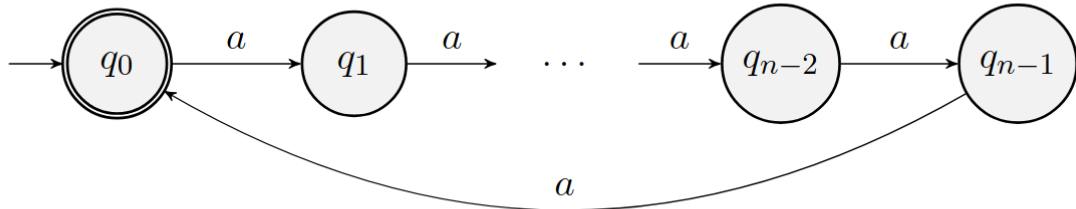
Q6

6) [0.5 points]

Let $B_n = \{a^k \mid k \text{ is a multiple of } n\}$. Show that for each $n \geq 1$, the language B_n is regular.

A6

$\forall n \in \mathbb{N}^+$, there exists a $NFA(N_n)$ that $L(N_n) = B_n$:



Hence, B_n is regular.

Note: it can also be proved by giving the regular expression $(a^n)^* = (\underbrace{aa \dots a}_n)^*$